❖ **Amenorrhea:** It is absence of menstruation.

❖ **Primary amenorrhea:** Menstruation does not begin at puberty.

❖ **Secondary amenorrhea:** Normal and regular menstrual periods which become increasingly abnormal and irregular or absent.

❖ **Dysmenorrhea**: It is characterized by severe and frequent menstrual cramps and pain associated with menstruation.

❖ **Primary Dysmenorrhea**: Women experience abnormal uterine contractions resulting from a chemical imbalance in the body. Mostly in adolescents

❖ **Secondary Dysmenorrhea:** It is caused by other medical conditions, most often endometriosis.

❖ **Menorrhagia:** It is the medical term for excessive menstrual bleeding. In a normal menstrual cycle, women on average lose about 30 ml of blood for about 7 days of menstruation. If bleeding exceeds 7 days or too heavy (over 80 ml), then it is called as menorrhagia. The main cause of menorrhagia is an imbalance of estrogen and progesterone in the body.

❖ **Abnormal Bleeding:** Abnormal vaginal bleeding (excluding menses)

- Bleeding between menstrual periods
- Bleeding after sex
- Bleeding after menopause

❖ **Metrorrhagia:** It refers to menstrual bleeding that is normal in amount but occurs of irregular intervals, between the menstrual periods hemorrhage from the uterus, independent of menstruation.

❖ **Menopause:** It is the process through which a woman ceases to be fertile.

5.3 FERTILIZATION AND FETAL DEVELOPMENT

Definition: It is the process during which a haploid male gamete (sperm) unites with a haploid female gamete (oocyte) to form a single cell (ZYGOTE). Is called fertilization.

The development of fetus divided in to 3 periods.

1. Pre-embryonic period (0 to 2 weeks)
2. Embryonic period (3-8 weeks)
3. Fetal period (9th week to birth of the baby)

1. Pre- embryonic period (0-2 Weeks)

- During coitus, sperm is released by male partner into the vagina of the female partner is called as insemination.
- The motile sperms swim and pass the cervix to enter into the uterus and finally to reach the ovum released by the ovary in the ampullary isthmic junction.
- Fertilization takes place in the ampulla-isthmic junction. Chemical signals from oocyte attract the sperms.
- The sperm after reaching the ovary in the ampullary isthmic junction comes in contact with the zona-pellucida layer of the ovum and block the entry of the additional sperms thus only one sperm fertilizes the ovum.
- The secretions of acrosome help the sperm to enter into the ovum through zonapellucida and the plasma membrane and thus secondary oocyte completes meiosis II and results in the formation of a second polar body and haploid ovum.
- The haploid nucleus of the sperm and ovum fuse together to form a zygote which develops into new individual.

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